Workshop Pole Barn: Truss Package

The brief for the truss plant · 32 × 48 ft gambrel attic trusses with a connected loft floor · Prairie, Elmore County, Idaho · planned for 2028

Every number here comes from the same design model as the rest of the package. Nothing is ordered or permitted yet; an Idaho-licensed engineer confirms every load and dimension.

What We Are Asking the Truss Plant For

1. **Design and seal 25 gambrel attic trusses with a connected loft floor** at 2′-0″ OC, numbered T1 at the west gable to T25 at the east: each bottom chord is a loft joist spanning the full 32′-0″ from wall line to wall line. 23 are common and 2 are gable-end (T1 and T25).
2. **Keep the loft room open and as tall as the shape allows:** no webs or posts in the 18′-0″ between the two pitch-break lines, with 7′-1″ clear at its edges, crossed only by a **collar tie** on T1 to T20 with its underside 8′-0″ above the loft floor (headroom). Brace the rest of the truss in the knee zones and with plates (bracing and blocking).
3. **Keep the roof shape:** the pitch break and the ridge stay put (the county's 27 ft peak limit leaves 8″ to spare), and the underside of the floor stays at 13′-2″ for the door below (profile).
4. **Design for 120 psf ground snow**, 115 mph wind, and a 40 psf loft floor (loads).
5. **Land on the carriers:** (3) 1¾″ × 9¼″ LVL bolted to the inside face of every post, the wall plate (bearing).
6. **Build in the five special cases:** the stair opening, the hoist beam, the loft doors, the loft window, and the gable-end trusses (special trusses).
7. **Send back** sealed drawings, the reactions, the clear headroom your design gives, and a quote with the boom-truck delivery as its own line (what to send back).

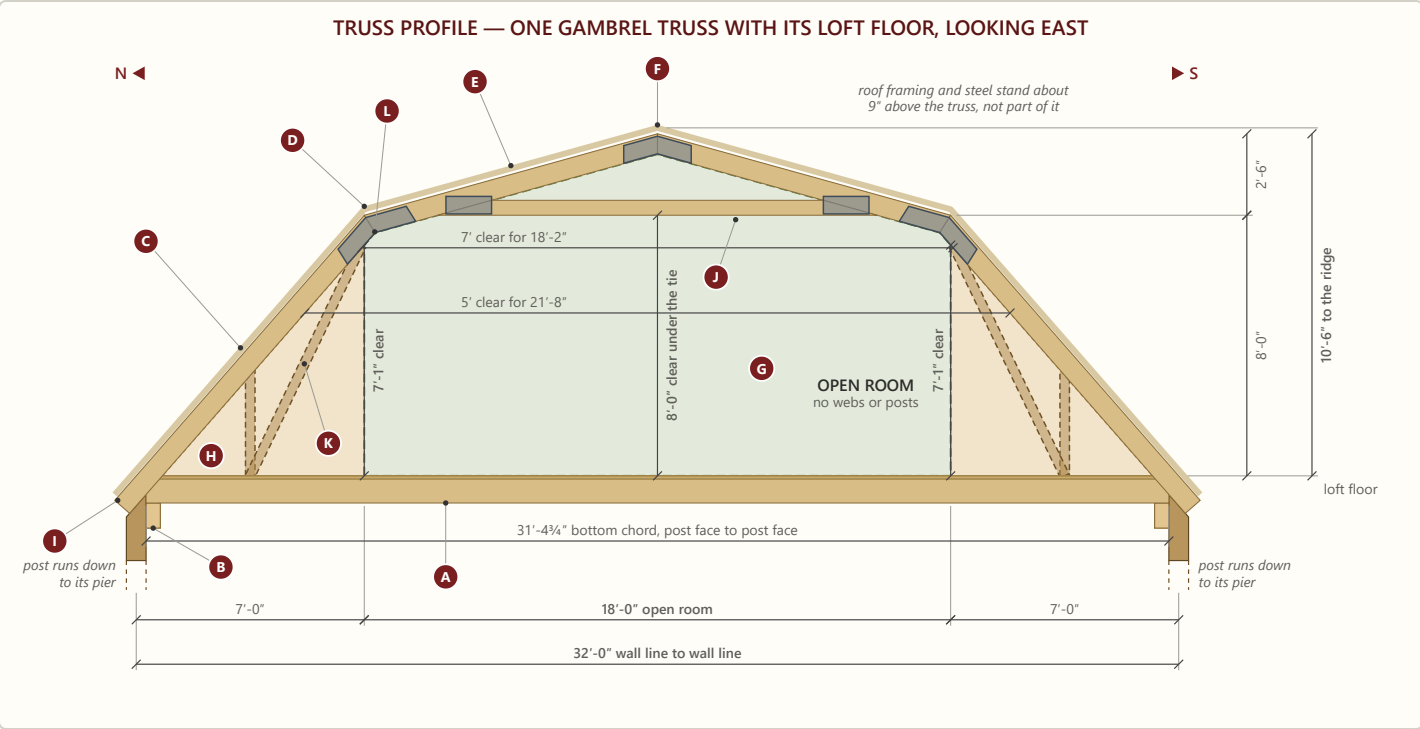
The Truss at a Glance

Item	What we need
Building	32 × 48 ft gambrel in Prairie, Elmore County, Idaho (about 4,750 ft): an RV garage and workshop below and a storage loft above. Sidewall posts stand at 8 ft.
Span	32′-0″ from wall line to wall line, the centers of the posts. The bottom chord is 31′-4¾″ long between the post faces and bears 5¼″ on each carrier; the clear span between the carriers is 30′-6¼″.
Bottom chord and floor	The chord is 9¼″ deep and carries ¾″ T&G plywood (laid by the builder), so the floor is 10″ thick. Its top is 14′-0″ above the main floor and its underside is fixed at 13′-2″: the 14′ × 12′ overhead door needs 12″ above it for its track, and this leaves 14″.
Top chords	2×8 (7¼″ deep) assumed; you set the real size. Lower slope 8′-0″ up in 7′-0″ (13.7:12); upper slope 2′-6″ up in 9′-0″ (3.3:12). Tail 1′-0″ along the slope past the wall line (8″ out). The heel is a 2×6 block between the chords with plates on both faces, **at least 12″** high at the bearing, for insulation depth (bearing).
Collar tie	One on each of T1 to T20, joining the two upper top chords with its underside 8′-0″ above the loft floor: 13′-6″ long between the chords, with a plate on both faces at each end. A 2×6 on edge is assumed; you size it and its connections. None on T21 to T25, where the hoist beam passes (bracing).
Roof on the trusses	2×6 purlins laid flat on the top chords at 16 to 24″ OC carry a ⅝″ OSB deck, underlayment, and 26 ga steel; the roof stands about 9″ above the trusses at the ridge.

Design Loads

Parameter	Value	Notes
Ground snow load (Pg)	**120 psf**	The design load. Elmore County's snow-load map puts Prairie in its 120 psf zone, where an Idaho engineer must design the building, and the county required a neighbor's building to be designed for it.
Flat-roof snow (approx.)	≈ 84 psf	0.7 × Pg for a heated, insulated building. You calculate the real figure: slope reductions, unbalanced snow, and the drift at the pitch break.
Wind and seismic	**115 mph**, Exposure C; county seismic zone C	ASCE 7, Risk Category II, an enclosed building on open ranch land; 115 mph is the county's residential figure. The engineer sets the seismic design category.
Loft floor	40 psf live load minimum, L/360 or stiffer	Quote **60 psf** as an alternate: the loft holds tools and stock. Tell us what deflection your design gives.
Dead loads (estimates)	Top chord 7 psf; bottom chord 10 psf	Steel, underlayment, OSB, purlins, and insulation on the top chord; plywood and floor covering on the bottom chord. Add the truss's own weight, and use your standard values where they are higher.
Concentrated load	Hoist beam, T21 to T25	A 500–1,000 lb working load at the beam's tip (S2); the engineer sends the reactions.

The Truss in Profile



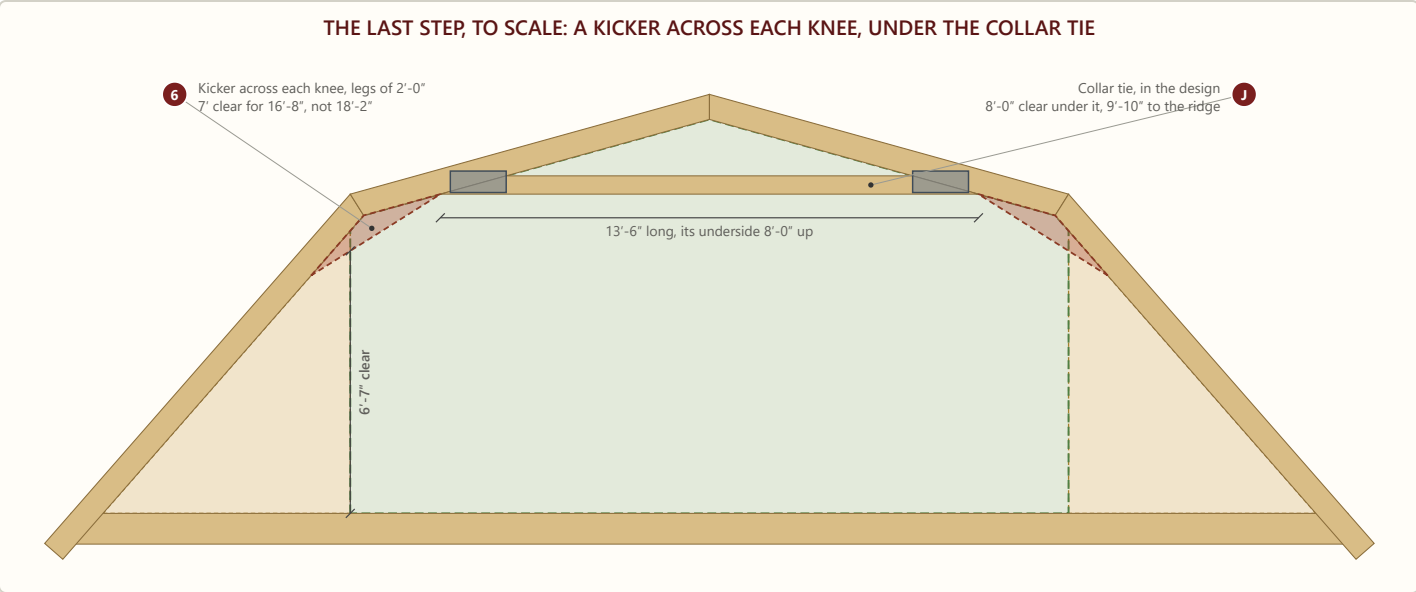
One truss across the 32′-0″ width, looking east, drawn to scale with 7¼″ top chords. The green zone is the open room; the amber zones are the knee zones, where the roof is too low to stand. The collar tie (J) joins the two upper top chords. The dashed members in the knee zones (K) are webs and kickers the plant may add, and the gray patches (L) are plates; all three are explained under Bracing and Blocking, next. The pitch break and the ridge are fixed; the plant sets the webs, the plates, the heel, and the tail.

Marker	What it shows
A	**Bottom chord and loft floor.** 9¼″ deep and 31′-4¾″ long between the post faces, with ¾″ T&G plywood on top. The underside is fixed at 13′-2″ above the main floor, and the top is the loft floor at 14′-0″. If the span needs a deeper floor, tell us the depth: the extra can only come out of the loft's headroom.
B	**Carrier (the wall plate).** (3) 1¾″ × 9¼″ LVL bolted to the inside face of each post; the bottom chord bears 5¼″ on it. The posts are 3-ply 2×8, 4½″ along the wall and 7¼″ across it, centered on the wall lines.
C	**Lower top chord.** 13.7:12, from the wall line up to the pitch break.
D	**Pitch break.** 7′-0″ in from the wall line and 8′-0″ above the loft floor. Fixed.
E	**Upper top chord.** 3.3:12, from the pitch break to the ridge.
F	**Ridge.** 16′-0″ in from each wall line and 10′-6″ above the loft floor. Fixed: the roof stands about 9″ above it, and the county's 27 ft peak limit leaves 8″.
G	**The open room.** The 18′-0″ between the two pitch-break lines, from the loft floor up to the underside of the top chords. No webs or posts. Clear height 7′-1″ at its edges and 9′-10″ at the ridge; the collar tie (J) caps the headroom at 8′-0″ across its 13′-6″.
H	**Knee zones.** The 7′-0″ outside each pitch-break line. Webs, kickers, plates, and blocking go here (K and L), and a web may stand in the plane of the pitch-break line itself.
I	**Heel and tail.** The roof line passes through the wall line at the loft floor level, and the tail runs 1′-0″ along the slope past it. The heel is a block between the chords with plates on both faces (see bearing), at least 12″ high at the outer edge of the bearing, so the roof line may rise near the wall for it. The pitch break and the ridge may not move.

Bracing and Blocking

The owner wants a **collar tie** across the upper slopes to help the roof carry the 120 psf snow, and it is in the design (J): it is the one member that crosses the open room. Besides it, the pitch break (the knee) is the joint that works hardest, so we expect bracing there. Webs, kickers, blocking, and plates cost the open room nothing when they stay in the knee zones or lie flat on the faces of the chords, so please use them first; the profile shows one way, marked K and L. Steps 1 to 6 are in the order we prefer them: send the first design that works, say which steps it uses, and price the later ones as alternates.

Step	What	Where	What it costs the open room
J	**Collar tie**, in the design	Between the two upper top chords on T1 to T20, its underside 8′-0″ above the loft floor, 13′-6″ long. A 2×6 on edge with a plate on both faces at each end is assumed: you size it and its connections. None on T21 to T25, where the hoist beam and its trolley pass: say which of steps 1 to 6 carry those five trusses.	Walking space is unchanged. The tallest headroom is 8′-0″, under the tie; above it the underside of the chords still rises to 9′-10″ at the ridge. If your design needs the tie lower or deeper, say what the headroom becomes.
1	**Webs and kickers**, with solid blocking between the trusses at the web lines and the pitch break	K The knee zones only. The profile's dashed members are a vertical web 3′-6″ in from each wall line and a kicker from its foot up to the pitch break: you place and size them, and may use your own layout.	Nothing in the room, blocking included. The floor plywood is cut around each web foot, so show where they land.
2	**Plates** at every joint, and a **gusseted knee**	L Flat steel plates on both faces of the chords at the pitch break and the ridge, large at the knee, with extra plies or LVL there if it needs them.	Nothing, while the underside of the chord stays where the profile puts it.
3	**Heavier chords**	More plies, or LVL, in the same depth.	Nothing in the room.
4	**Closer trusses**	24″ OC becomes 16″ OC, or the trusses are doubled.	Nothing in the room. The layout, the purlins, and the carriers change.
5	**Deeper chords**	Top chords deeper than 7¼″.	About an inch of headroom for each inch of depth, and more at the pitch break.
6	**A short kicker across each knee**	Inside the room's top corners, with legs of up to 2′-0″ along each chord from the inside of the pitch break.	The clear height at the room's edge falls from 7′-1″ to 6′-7″, and the width with 7′ of clear height from 18′-2″ to 16′-8″.



The last step, to scale: the short kicker across each knee (6), which cuts the room's top corners along a straight line between the ends of its legs, drawn under the collar tie (J) that is in the design. The numbers are in the table above. List the permanent lateral bracing and bridging you need on the placement plan.

Headroom and Walking Space

The loft is a room people walk and work in, and the profile gives the headroom this roof shape allows under the owner's collar tie. We ask the plant to protect it.

- **Clear height** is measured from the top of the loft floor up to the **underside** of the top chords, and under the collar tie to the underside of the tie. The plans page measures headroom to the top of the chords (the roof line), which is 7½″ higher at the ridge and 11″ higher at the pitch break.
- **The roof line is fixed.** Do not raise the ridge or the pitch break. The roof line may rise near the wall for the heel.
- **The clear heights below are the least we accept**, measured under 7¼″ top chords. If your loads need deeper chords, tell us how many inches deeper and what the clear heights become: each extra inch of chord costs about an inch of headroom, and more at the pitch break.
- **The room stays open** between the pitch-break lines: no webs or posts. The collar tie on T1 to T20 and the hoist beam at the east end (S2) are the only things in it, so the tallest headroom is 8′-0″, under the tie. The rest of the bracing goes in the knee zones and on the faces of the chords; only the last step in bracing and blocking comes into the room.
- **If the open room cannot be built** at 24″ OC for these loads, send the closest design that can, and say what it gives up. The steps in bracing and blocking are in the order we prefer them. Price each one.

Headroom in from a wall line

Both heights are above the loft floor: the roof line height is to the top of the chords, the clear height to their underside. On T1 to T20 the collar tie caps the clear height at 8′-0″ from 9′-3″ to 22′-9″ in.

In from a wall line	Roof line height	Clear height
1′-0″	1′-2″	0′-3″
2′-0″	2′-3″	1′-4″
3′-0″	3′-5″	2′-6″
4′-0″	4′-7″	3′-8″
5′-0″	5′-9″	4′-10″
6′-0″	6′-10″	5′-11″
7′-0″ (pitch break)	8′-0″	7′-1″
8′-0″	8′-3″	7′-8″
10′-0″	8′-10″	8′-2″
12′-0″	9′-5″	8′-9″
14′-0″	9′-11″	9′-4″
16′-0″ (ridge)	10′-6″	9′-10″

18′-2″ of the loft width has **7′ or more** of clear height, and 21′-8″ has **5′ or more**.

Open room outline (under the chords)

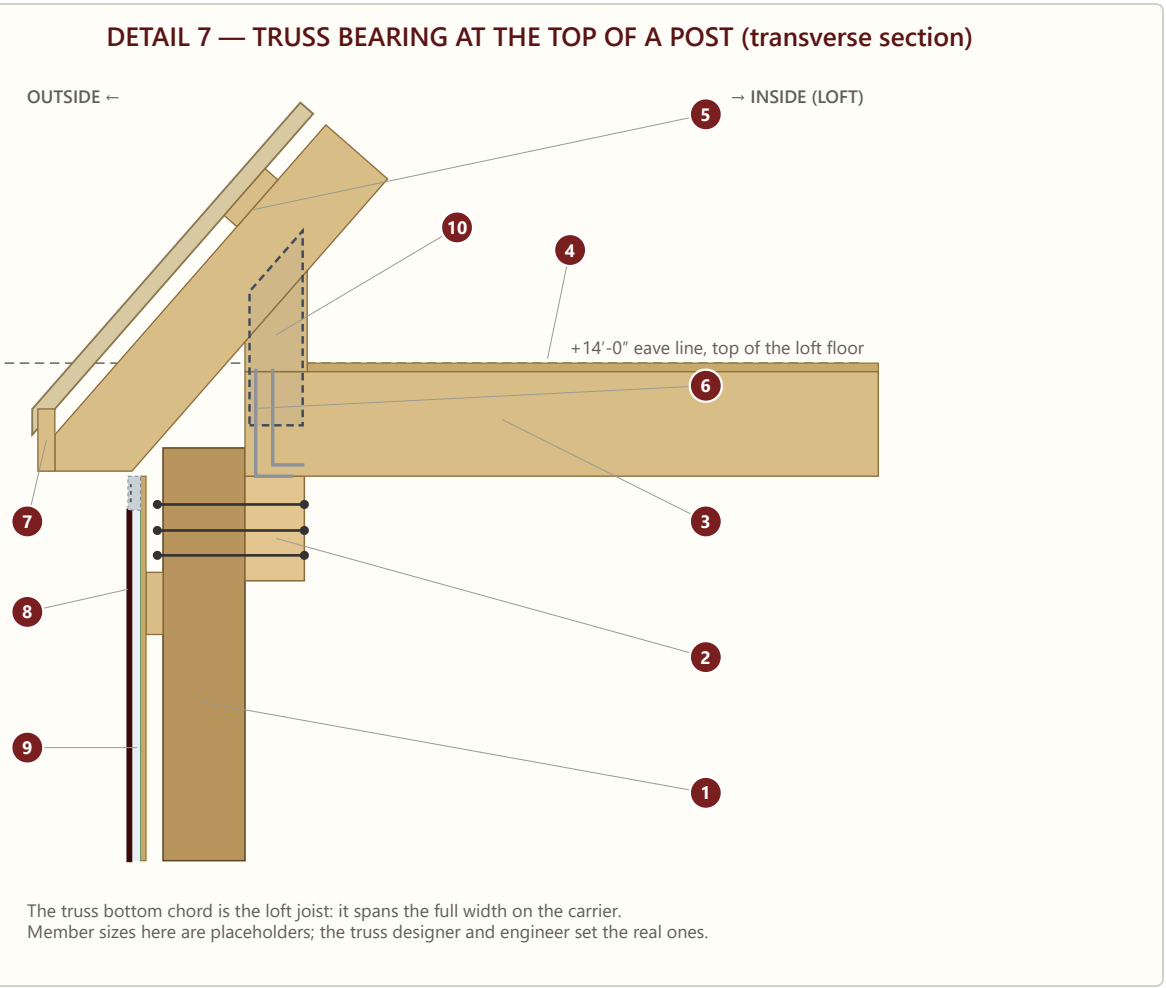
Open room point	In from the north wall line	Clear height
Chord underside meets the floor	0′-10″	0′-0″
Room edge, on the pitch-break line	7′-0″	7′-1″
Underside at the pitch break	7′-4″	7′-6″
Collar tie underside, to 22′-9″	9′-3″	8′-0″
Ridge	16′-0″	9′-10″

Roof line (top of the chords)

Roof line point	In from the north wall line	Above the loft floor
Tail end	8″ outside the wall line	9″ below
Wall line	0′-0″	0′-0″
Pitch break	7′-0″	8′-0″
Ridge	16′-0″	10′-6″

The south half mirrors the north. The open room's outline is the roof line moved down 7¼″, measured square to the slope. The roof line may rise near the wall for the heel.

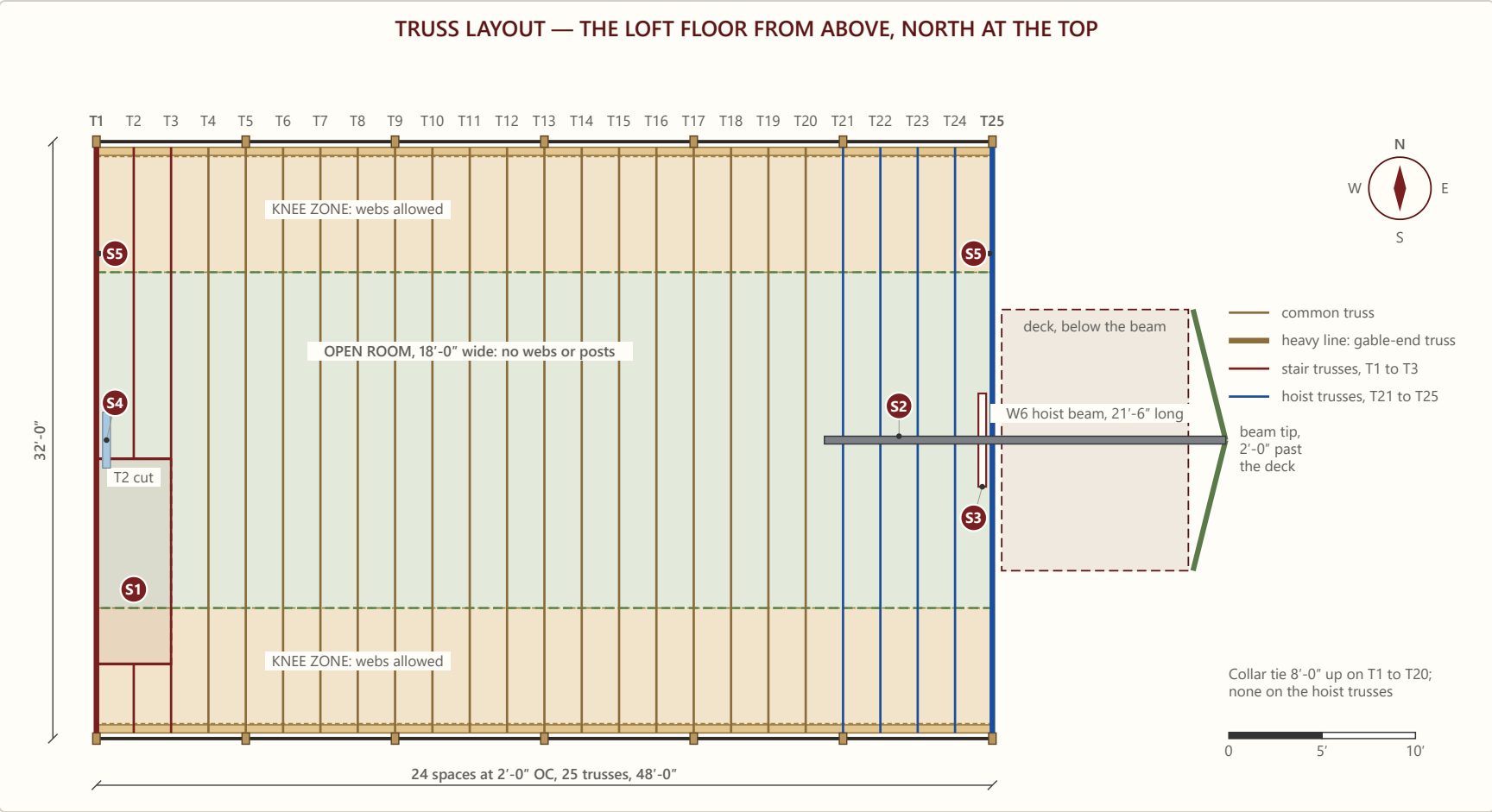
Bearing and Heel



A transverse section through one sidewall post at the eave. The truss's bottom chord is the loft joist: it spans the full width with no bearing between the walls and lands on a (3) 1¾″ × 9¼″ LVL carrier side-bolted to the inside face of every post. The heel is a 2×6 block between the top chord and the bottom chord at the bearing, with steel plates on both faces (dashed: behind the cut); the plant sizes the block and the plates. The tail's shape is a placeholder, within the limits in the profile.

Marker	What it is
1	3-ply 2×8 laminated post (4½″ × 7¼″) on a wet-set bracket below
2	(3) 1¾″ × 9¼″ LVL truss carrier, side-bolted to the post with (3) ½″ galvanized through-bolts per post
3	Truss bottom chord (2×10, a placeholder size), bearing on the carrier and acting as the loft joist across the span
4	¾″ T&G plywood loft floor, long edge across the chords
5	Truss top chord (2×8, a placeholder size), the lower gambrel slope rising to the pitch break
6	H2.5A hurricane ties, 2 per chord bearing, one on each side
7	2×6 sub-fascia closing the plumb-cut end of the truss tail (about 12″ of overhang), with the eave trim wrapped around it
8	Wall steel stops under the vented closure strip that crosses the top of the rainscreen cavity
9	Wall assembly outside the post: 2×6 girt, OSB, wrap, rainscreen, and steel
10	Heel block (a 2×6 on end) between the top chord and the bottom chord at the bearing, with the plant's steel plates on both faces (dashed: behind the cut). The loft floor butts to the block. The heel stands 13″ above the floor at the bearing's outer edge, and at least 12″

Special Trusses and Conditions



The loft floor from above, north at the top. T1 is the west gable truss and T25 the east gable truss, 2′-0″ apart. **S1** stair opening, **S2** hoist beam, **S3** loft doors, **S4** loft window, **S5** gable-end trusses: the next section explains each. Red trusses are the stair trusses and blue ones carry the hoist beam; T1 to T20 each carry a collar tie. The green zone is the open room and the amber strips are the knee zones.

What Each Mark Asks For

Mark	Where	What the truss plant must allow for
S1	**Stair opening.** T1 to T3, the first 4′-0″ from the west end	A 4′ × 11′ opening in the loft floor, from 17′-0″ to 28′-0″ south of the north wall line, over the second flight of the stair. Our drawings cut the bottom chord of T2 between two (2) 2×10 LVL headers, with T1 and T3 as the edges. The cut truss must still carry its roof and tie its heels: show how, or propose another arrangement (a girder truss at T3 with T2 left out, say) and what it does to the purlin span. Size T1 and T3 for the header reactions.
S2	**Hoist beam.** T21 to T25, from 39′-0″ east of the west end to the east gable	A W6×12 steel beam, 21′-6″ long, on the ridge line: 9 ft inside the loft and 12′-6″ out through the east gable, 2 ft past the deck's edge. Its centerline is 9 ft above the loft floor, so its top stops 7½″ below the underside of the top chords at the ridge; blocking fills that gap and bolts to the top chords. A 500–1,000 lb working load acts at its tip, shared with two braces to the deck. The engineer sends the reactions. Tell us what the ridge joints and chords of T21 to T25 can take, and any plates or blocking you need. These five trusses take no collar tie, because the trolley and chain pass under the beam at that height: say how you carry their upper chords without one (steps 1 to 6 of bracing and blocking).
S3	**Loft doors.** T25, the east gable truss	5′ × 6′ double doors, rough opening 5′-6″ × 6′-6″, centered on the gable (13′-3″ to 18′-9″ from the north wall line), with the bottom at the loft floor and a (2) 2×10 LVL header above. Keep the webs, studs, and bracing of the gable truss out of the opening up to 7′-3″ above the loft floor.
S4	**Loft window.** T1, the west gable truss	A 3′ × 4′ single-hung window, centered on the gable (14′-6″ to 17′-6″ from the north wall line), with its sill 3′ above the loft floor. Keep the webs and studs of T1 out of it.
S5	**Gable-end trusses.** T1 and T25	Solid-faced, with vertical studs at 24″ OC for the siding, and outlookers or an equal for the 1′-0″ roof overhang at each gable. They sit on the end walls as well as on the carriers: for the quote, assume bearing along the wall, except across the 14 ft door at the east end, where a header carries it. The engineer sends the bearing conditions.

What to Send Back

1. **Sealed truss drawings** by an Idaho-licensed engineer, for each truss type: common, gable-end, the stair truss (T2), and the hoist trusses (T21 to T25), with member sizes, lumber grades, plate sizes, and the loads you used.
2. **A placement plan** with the web layout, the permanent lateral bracing and bridging you require (including the solid blocking between trusses and any bracing the collar ties need), and the purlin spacing your top chords assume.
3. **Reactions**, vertical and uplift, at every bearing, for each truss type, so the engineer can size the carriers, their bolts, the posts, and the hoist anchors, plus the hurricane ties you specify (we assume H2.5A, 2 per bearing).
4. **The clear headroom your design gives**: at the room's edge, under the collar tie, at the ridge, and the room width at 7′ and 5′ clear. If your top chords are deeper than 7¼″, say how much. Say which steps of bracing and blocking your design uses.
5. **The collar tie you design** for T1 to T20: its size and grade, the end plates and connections, the pull it carries under the snow, and how you carry the upper chords of T21 to T25, which take none.
6. **The heel block and plates you propose** (we draw a 2×6 block), the tail, and the post-top cut they need.
7. **The snow cases you used**: drift and unbalanced snow at the pitch break.
8. **A quote by line**: common trusses with their collar ties, gable-end trusses, the special trusses (T2 and T21 to T25), any connectors you supply, and the alternates: a 60 psf loft floor, 16″ OC, and any step of bracing and blocking beyond the knee zones (the kicker across the knee).
9. **Delivery by boom truck, as its own line**: the delivery fee, the boom-truck time and the hours included on site, and whether the driver can set the bundles on the walls (the walls stand 14′-0″ above the main floor). We will confirm the route onto the pad before the delivery date.
10. **Lead time** from signed drawings to delivery, and how long the quote holds.
11. **Anything you cannot meet**: name the requirement and propose the closest design.

Mountain Prairie Acres · Workshop pole barn truss package · updated October 23, 2026 · drawn from `docs/workshop_model.js`.